QQ-A-250/14E
January 28, 1971
SUPERSEDING
Fed. Spec. QQ-A-250/14D
March 17, 1967

FEDERAL SPECIFICATION

ALUMINUM ALLOY 7178, PLATE AND SHEET

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

(This specification forms a part of the latest issue of Federal Specification QQ-A-250).

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the specific requirements for aluminum alloy 7178 plate and sheet material; the general requirements are covered in QQ-A-250.

1.2 Classification.

1.2.1 Tempers. The plate and sheet shall be classified as O, T6, T62, T651, or F temper, as specified (see 6.2). The definitions of these tempers shall be as specified in American National Standard ANSI H35.1.

2. APPLICABLE DOCUMENTS

Latest issue of Federal Specification:

QQ-A-250 - Aluminum and Aluminum Alloy Plate and Sheet; General Specification for.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC. 20402.

FSC 9535

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(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

American National Standards Institute ANSI Standard:

H35.1 - Alloy and Temper Designation Systems for Aluminum.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, New York 10018.)

3. REQUIREMENTS

3.1 Chemical composition.

3.1.1 The chemical composition shall conform to the requirements specified in table I.

Table I. Chemical composition 1/

	Percent	
	Minimum	Maximum
Zinc	6.3	7.3
Magnesium	2.4	3.1
Copper	1.6	2.4
Chromium	0.18	0.35
Iron	-	.50
Silicon	-	.40
Manganese	-	.30
Titanium	-	.20
Others, each	-	.05
Others, total	-	.15
Aluminum	Remainder	

1/ Analysis shall regularly be made only for the elements specified in table I. If, however, the presence of other elements is indicated in the course of routine analysis, further analysis shall be made to determine conformance to the limits specified for other elements.

3.2 Mechanical properties.

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3.2.1 Mechanical properties of material as supplied. The mechanical properties perpendicular to the direction of final rolling, except for material less than 9 inches in width, shall conform to the requirements of table II for the temper specified. For material less than 9 inches in width, the mechanical properties parallel to the direction of final rolling shall conform to the requirements of table II for the temper specified.

TABLE II. Mechanical properties

Temper	Thickness	Tensile strength, minimum	Yield strength at 0.2 percent offset, minimum	Elongation in 2 in. or 4 times D 1/, 2/, minimum
	Inches	P.s.i.	P.s.i.	Percent
O	0.015 thru 0.499	3/ 40,000	3/ 21,000	10
	0.500	3/ 40,000	-	10
T6 and T62 4/	0.015 thru 0.044	83,000	72,000	7
	.045 thru .249	84,000	73,000	8
T651 and T62 4/	0.250 thru 0.499	84,000	73,000	8
	.500 thru 1.000	84,000	73,000	6
	1.001 thru 1.500	84,000	73,000	4
	1.501 thru 2.000	80,000	70,000	3
F	All	5/	5/	5/

1/ Not required for material 1/2 inch or less in width.
2/ D represents specimen diameter.
3/ Maximum.
4/ Material in the T62 temper is not available from the materials producers.
5/ No requirements.

3.2.2 Mechanical properties after heat treatment. In addition to conforming to the requirements of 3.2.1, material in the annealed (O) and the as-fabricated (F) tempers shall, after proper solution heat-treatment and aging, also conform to the requirements of table II for the T6 and T62 tempers. Material as received in the T6 and T651 tempers shall, after proper re-solution heat-treatment and aging, be capable of conforming to the requirements specified in table II for the T6 and T62 tempers.

3.3 Internal defects. When specified (see 6.2), plate shall be ultrasonically inspected (see QQ-A-250). Acceptance limits shall be as specified in table III.

TABLE III. Ultrasonic discontinuity acceptance limits 1/

Thickness	Maximum weight per piece	Discontinuity class 2/
Inches	Pounds	
0.500 thru 1.499	2,000	B
1.500 thru 2.000	2,000	A

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1/ Discontinuities in excess of those listed in table III may be allowed subject to the approval of the procuring activity, if it is established that they will be removed by machining or that they are in non-critical areas.
2/ The discontinuity class limits are defined in MIL-I-8950.

3.4 Marking. In addition to the marking required in Fed. Std. No. 184, material in the T6 and T651 tempers shall be identified by a lot number marked in at least one location on each piece.

4. QUALITY ASSURANCE PROVISIONS
(See QQ-A-250).

4.1 Mechanical tests after heat treatment.

4.1.1 Number of tests after heat treatment. From material in the annealed (O) and as-fabricated (F) tempers, an additional number of specimens equal to that required in QQ-A-250 shall be taken and tested after solution heat treatment and artificial aging to determine compliance with 3.2.2.

5. PREPARATION FOR DELIVERY
(See QQ-A-250.)

6. NOTES

6.1 Intended use. The material covered by this specification is intended for use in high-strength applications. This material has poor resistance to fracture propagation. This factor should be considered in the design of structures for weapons systems.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.

(b) Form and temper required (see 1.2.1).

(c) Dimensions required.

(d) Requirements for sizes not specifically covered (see QQ-A-250).

(e) Whether ultrasonic inspection of plate is required (see 3.3).

(f) Selection of applicable levels of preservation, packaging, and packing required if other than level C (see QQ-A-250).

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6.3 Temper. Temper T651 is available in plate only.

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